

```
In [1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [2]: df=pd.read_csv('Iris.csv')
df.head()
```

```
Out[2]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa

```
In [3]: df.shape
```

```
Out[3]: (150, 6)
```

```
In [4]: df.describe()
```

```
Out[4]:
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
In [5]: df.isnull().sum()
```

```
Out[5]: Id          0
SepalLengthCm      0
SepalWidthCm       0
PetalLengthCm      0
PetalWidthCm       0
Species           0
dtype: int64
```

```
In [6]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Id              150 non-null   int64
 1   SepalLengthCm   150 non-null   float64
 2   SepalWidthCm    150 non-null   float64
 3   PetalLengthCm   150 non-null   float64
 4   PetalWidthCm    150 non-null   float64
 5   Species         150 non-null   object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

```
In [7]: df.dtypes
```

```
Out[7]: Id              int64
SepalLengthCm         float64
SepalWidthCm          float64
PetalLengthCm         float64
PetalWidthCm          float64
Species              object
dtype: object
```

```
In [8]: df['Species'] = df['Species'].astype('category')
df
```

Out[8]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

150 rows × 6 columns

In [9]: `df = df.drop(columns=['Id'])`
In [10]: `from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler()
df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']] = scaler.fit
df[['SepalLengthCm', 'SepalWidthCm', 'PetalLengthCm', 'PetalWidthCm']]
)
df=df.round(2)`

In [11]: `from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
df['Species_Encoded'] = le.fit_transform(df['Species'])
df`

Out[11]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Species_I
0	0.22	0.62	0.07	0.04	Iris-setosa	
1	0.17	0.42	0.07	0.04	Iris-setosa	
2	0.11	0.50	0.05	0.04	Iris-setosa	
3	0.08	0.46	0.08	0.04	Iris-setosa	
4	0.19	0.67	0.07	0.04	Iris-setosa	
...	...	...	...	...	...	...
145	0.67	0.42	0.71	0.92	Iris-virginica	
146	0.56	0.21	0.68	0.75	Iris-virginica	
147	0.61	0.42	0.71	0.79	Iris-virginica	
148	0.53	0.58	0.75	0.92	Iris-virginica	
149	0.44	0.42	0.69	0.71	Iris-virginica	

150 rows × 6 columns



In [12]: df.head()

Out[12]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species	Species_Enc
0	0.22	0.62	0.07	0.04	Iris-setosa	
1	0.17	0.42	0.07	0.04	Iris-setosa	
2	0.11	0.50	0.05	0.04	Iris-setosa	
3	0.08	0.46	0.08	0.04	Iris-setosa	
4	0.19	0.67	0.07	0.04	Iris-setosa	



In []: